

Open dMRI dataset (no credentials required)

Stanford HARDI — 160 volumes, $b = 2000$

Sherbrooke 3-shell — 193 volumes, $b = 1000 / 2000 / 3500$

Brain extraction compared

each tool on the input it is designed for

FSL

bet

← mean $b = 0$

MRtrix3

dwi2mask

← full DWI series

DIPY

median_otsu

← full series, $b = 0$ idx

Dice similarity coefficient
between each pair

Table 2 · Figure 4

Tensor fitting compared

identical input; estimator varied deliberately

held identical across every arm

one brain mask (median_otsu) · one volume subset
 $b = 0$ + a single non-zero shell · no preprocessing

FSL

dtifit -wls
measured

MRtrix3

dwi2tensor
predicted

MRtrix3

-iter 0
measured

DIPY

TensorModel
predicted

FA and MD maps, one set per arm

physically admissible voxels only

FA in $[0, 1]$ $0 < MD \leq 3.0 \times 10^{-3} \text{ mm}^2/\text{s}$

Pearson r · Spearman ρ · MAE
Bland-Altman bias and 95% limits

Tables 2-4, 6, 7 · Figure 3

validated against a phantom with known eigenvalues

Table 5